

Project Overview

Objective:

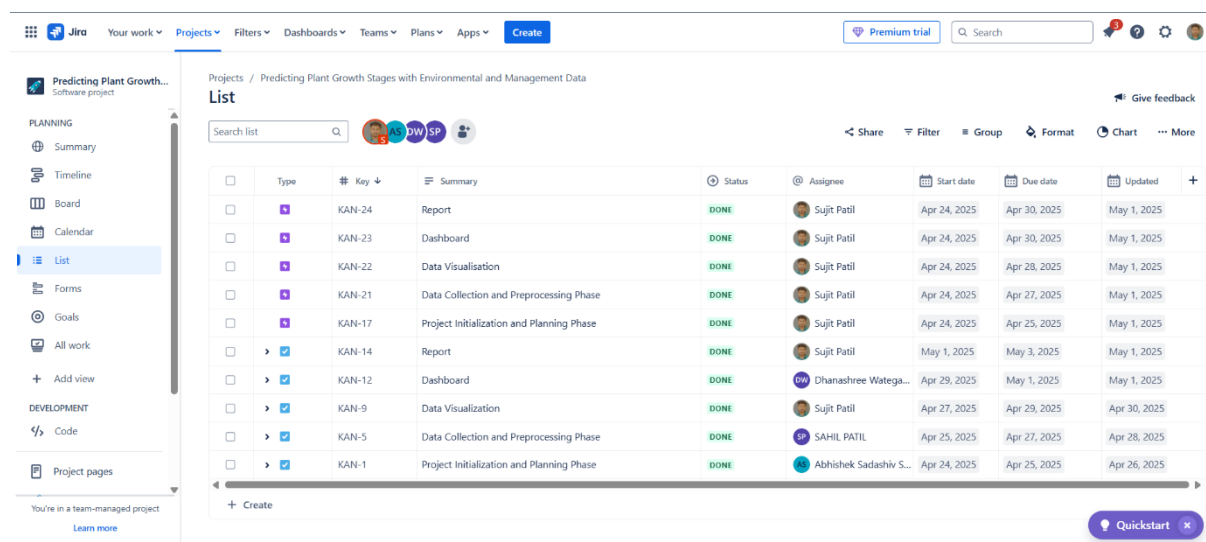
We aim to predict plant growth stages using environmental and farm management data. This will help farmers and agricultural experts make smarter, timely decisions for better crop yield and quality.

Scope:

To visualise how factors like temperature, rainfall, soil, and farm activities affect plant growth. It will support selected crops and regions based on available data and offer stage-wise predictions to guide field operations.

Jira:

The Project task is assigned to the members are as below in the screenshot



Type	Key	Summary	Status	Assignee	Start date	Due date	Updated
	KAN-24	Report	DONE	Sujit Patil	Apr 24, 2025	Apr 30, 2025	May 1, 2025
	KAN-23	Dashboard	DONE	Sujit Patil	Apr 24, 2025	Apr 30, 2025	May 1, 2025
	KAN-22	Data Visualisation	DONE	Sujit Patil	Apr 24, 2025	Apr 28, 2025	May 1, 2025
	KAN-21	Data Collection and Preprocessing Phase	DONE	Sujit Patil	Apr 24, 2025	Apr 27, 2025	May 1, 2025
	KAN-17	Project Initialization and Planning Phase	DONE	Sujit Patil	Apr 24, 2025	Apr 25, 2025	May 1, 2025
	KAN-14	Report	DONE	Sujit Patil	May 1, 2025	May 3, 2025	May 1, 2025
	KAN-12	Dashboard	DONE	Dhanashree Watega...	Apr 29, 2025	May 1, 2025	May 1, 2025
	KAN-9	Data Visualization	DONE	Sujit Patil	Apr 27, 2025	Apr 29, 2025	Apr 30, 2025
	KAN-5	Data Collection and Preprocessing Phase	DONE	SAHIL PATIL	Apr 25, 2025	Apr 27, 2025	Apr 28, 2025
	KAN-1	Project Initialization and Planning Phase	DONE	Abhishek Sadashiv S...	Apr 24, 2025	Apr 25, 2025	Apr 26, 2025

Problem Statement Creation

- **For a Small-Scale Farmer:**

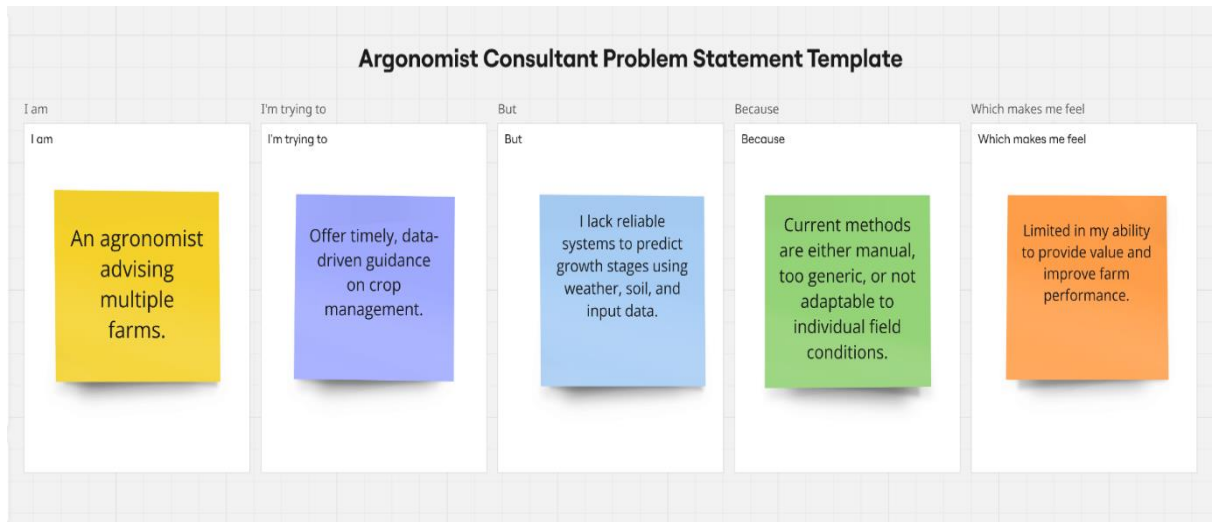
I am a small-scale farmer growing vegetables. I am trying to improve my crop yields by understanding plant growth stages better, but I struggle to predict when plants will transition between growth stages because I don't have access to accurate data or tools that analyze environmental and management factors. This makes me feel uncertain and frustrated about my farming decisions.

Farmer Problem Statement Template

I am	I'm trying to	But	Because	Which makes me feel
I am A small-scale farmer growing vegetables.	I'm trying to Improve my crop yields by understanding plant growth stages better.	But I struggle to predict when plants will transition between growth stages.	Because I don't have access to accurate data or tools that analyze environmental and management factors.	Which makes me feel Uncertain and frustrated about my farming decisions.

- **For an Agronomist:**

I am an agronomist advising multiple farms. I am trying to offer timely, data-driven guidance on crop management, but I lack reliable systems to predict growth stages using weather, soil, and input data because current methods are manual and too generic. This makes me feel limited in providing real value.



- **For a Researcher:**

I am a researcher working on agricultural technology. I am trying to build predictive models for plant growth to support precision agriculture, but I find it hard to integrate diverse environmental and management data sources effectively because no standardized system exists. This makes me feel stuck and delayed in my innovation process.

Researcher / AgTech Developer Problem Statement Template				
I am	I'm trying to	But	Because	Which makes me feel
I am	I'm trying to	But	Because	Which makes me feel
A researcher working on agricultural technology.	Build predictive models for plant growth to support precision agriculture.	I find it hard to integrate diverse environmental and management data sources effectively.	There is no standardized or streamlined system to capture and process all relevant data.	Stuck and delayed in my innovation process.

• Table Creation of Various Problem Statement

Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-1	A small-scale farmer growing vegetables.	Improve my crop yields by understanding plant growth stages better.	I struggle to predict when plants will transition between growth stages	I don't have access to accurate data or tools that analyse environmental and management factors	Uncertain and frustrated about farming decisions
PS-2	An agronomist advising multiple farms.	Offer timely, data-driven guidance on crop management.	I lack reliable systems to predict growth stages using weather, soil, and input data.	Current methods are either manual, too generic, or not adaptable to individual field conditions.	Limited in my ability to provide value and improve farm performance.

Problem Statement (PS)	I am	I'm trying to	But	Because	Which makes me feel
PS-3	A researcher working on agricultural technology.	Build predictive models for plant growth to support precision agriculture.	I find it hard to integrate diverse environmental and management data sources effectively.	There is no standardized or streamlined system to capture and process all relevant data.	Stuck and delayed in my innovation process.

Reference: <https://miro.com/templates/customer-problem-statement/>

Project Proposal (Proposed Solution) template

This project proposal outlines a solution to address a specific problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the approach, key features, and resource requirements, including hardware, software, and personnel.

Project Overview	
Objective	We aim to predict plant growth stages using environmental and farm management data. This will help farmers and agricultural experts make smarter, timely decisions for better crop yield and quality.
Scope	To visualise how factors like temperature, rainfall, soil, and farm activities affect plant growth. It will support selected crops and regions based on available data and offer stage-wise predictions to guide field operations.
Problem Statement	
Description	Farmers often follow a general schedule for crop management, which may not match real-time plant growth. This leads to missed opportunities and inefficiencies.
Impact	Mistimed actions can lower crop output, waste resources, and increase risks. A data-driven approach can solve this and boost productivity.
Proposed Solution	
Approach	We will gather and analyze environmental and management data, use prediction models to estimate growth stages, visualize everything in Power BI and

	use filters and visuals (like charts, bar graphs, heat maps and slicers) for deeper insights. The dashboard will be interactive and user-friendly
Key Features	Visual timeline of plant growth stages Charts showing impact of weather and management actions Predictive insights for better planning Region, crop, and time filters

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	2 x NVIDIA V100 GPUs
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	1 TB SSD
Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn, pandas, numpy

Development Environment	IDE, version control	Jupyter Notebook, Git
Data		
Data	Source, size, format	Kaggle dataset, 1,000 images

Data Exploration and Preprocessing Template

Identifies data sources, assesses quality issues like missing values and duplicates, and implements resolution plans to ensure accurate and reliable analysis.

Section	Description
Data Overview	The dataset contains 100 records capturing plant growth stages, environmental factors (temperature, humidity, soil moisture), and management activities (fertilizer used, water given). The goal is to predict growth stages based on these features.
Data Cleaning	Checked and handled missing values by applying mean/mode imputation where necessary. Verified and removed duplicate records. Corrected minor inconsistencies in text fields (e.g., Growth Stage labels).
Data Transformation	Used Power Query for filtering irrelevant records, sorting data by Date and Growth Stage, pivoting tables for analysis, and creating calculated columns like "Management Effort" (sum of water and fertilizer inputs).
Data Type Conversion	Converted Date fields to proper Date/Time format, numeric fields (Temperature,

	Humidity, Soil Moisture) to Decimal format, and Growth Stage fields to Text format.
Column Splitting and Merging	Split Date fields into Month and Year for seasonal analysis. Merged Water and Fertilizer columns to create a "Total Management Effort" metric for better insights.
Data Modeling	Defined logical relationships among fields (e.g., Plant ID to Growth Stage progression). Created basic DAX measures for Average Temperature, Average Humidity, and Total Water Usage.
Save Processed Data	Saved the cleaned and transformed dataset into a new Excel file, ensuring it is ready for visualization and analysis in Power BI.

Data Quality Report Template

The Data Quality Report Template will summarize data quality issues from the selected source, including severity levels and resolution plans. It will aid in systematically identifying and rectifying data discrepancies.

Data Source	Data Quality Issue	Severity	Resolution Plan
Dataset 1: Plant Growth Data Classification (Kaggle)	Minor random noise in environmental readings (Temperature, Humidity).	Low	Apply smoothing techniques such as Moving Average Filter to stabilize environmental fluctuations.
Dataset 2: Greenhouse Plant Growth Metrics (Kaggle)	Inconsistent text casing in Growth Stage labels (e.g., "flowering" vs "Flowering").	Low	Use Power Query Text Functions (Text.Proper) to standardize text casing.
Dataset 1: Plant Growth	Slight class imbalance	Moderate	Apply sampling

Data Classification (Kaggle)	among different Growth Stages.		techniques or use stratified models to ensure balanced analysis and avoid bias.
Dataset 3: Custom Synthetic Dataset	Risk of incomplete records if manual generation errors occur.	Low	Validate generated records using consistency rules and automate data validation scripts.
Dataset 2: Greenhouse Plant Growth Metrics (Kaggle)	Minor missing values in environmental columns.	Low	Use imputation techniques such as mean or median substitution to fill missing values without biasing the dataset.

Data Collection Plan & Raw Data Sources Identification Template

Elevate your data strategy with the Data Collection plan and the Raw Data Sources report, ensuring meticulous data curation and integrity for informed decision-making in every analysis and decision-making endeavor.

Data Collection Plan Template

Section	Description
Project Overview	This project aims to predict plant growth stages using environmental parameters (temperature, humidity, soil moisture) and farm management practices (fertilizer usage, water given). The goal is to help improve agricultural decision-making with better stage prediction.
Data Collection Plan	Data has been collected from publicly available agricultural datasets, research articles, and synthetically generated datasets based on real-world environmental factors.
Raw Data Sources Identified	1. Kaggle datasets related to Plant Growth and Environment.2. Research articles and agronomy studies.3. Synthetic dataset created by the team to simulate controlled plant growth environments.

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Raw Data Sources Template

Source Name	Description	Location/URL	Format	Size	Access Permissions
Dataset 1	Dataset containing plant growth stages with environmental parameters like temperature, humidity, and soil moisture, used for classification	Plant Growth Data Classification (Kaggle)	CSV	50 MB	Public
Dataset 2	Greenhouse plant growth metrics dataset with over 30,000 samples including chlorophyll levels, biomass data, and growth classification.	Greenhouse Plant Growth Metrics (Kaggle)	CSV	100 MB	Public

Dataset 3	Custom dataset created by the project team simulating real-world environmental and management parameters to fill data gaps and enhance modeling.	Locally Created Dataset (Internal Team Resource)	Excel	10 MB	Private
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Business Question and Visualization Report

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data

visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting.

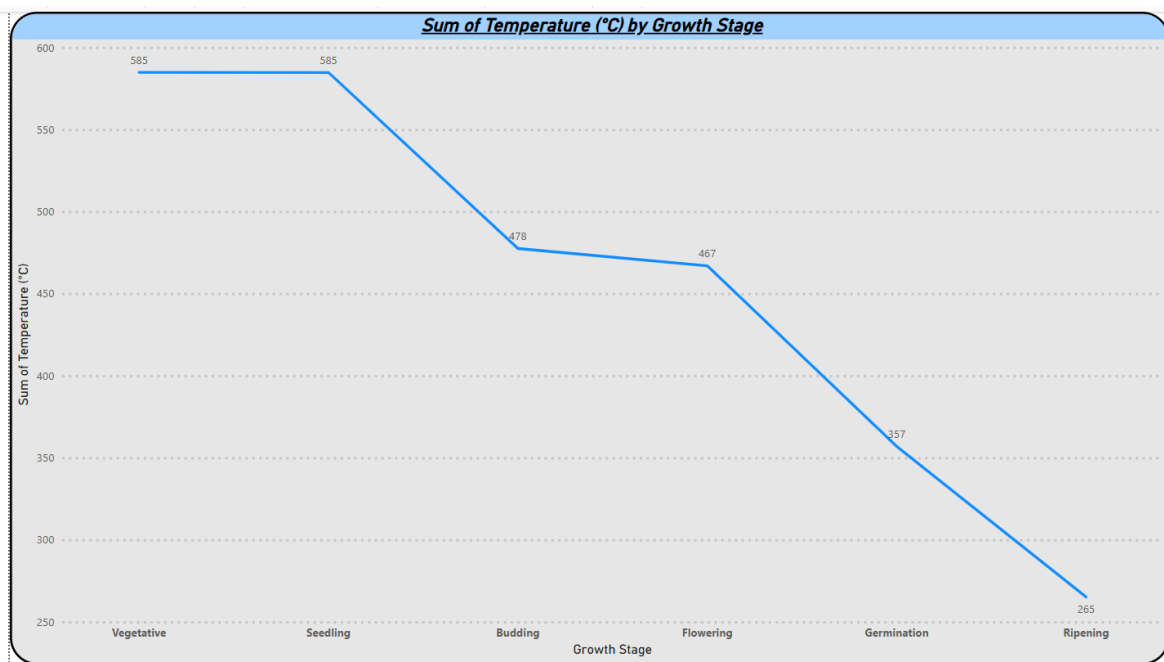
Business Questions and Visualisation

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Power BI. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

➤ Visualisations

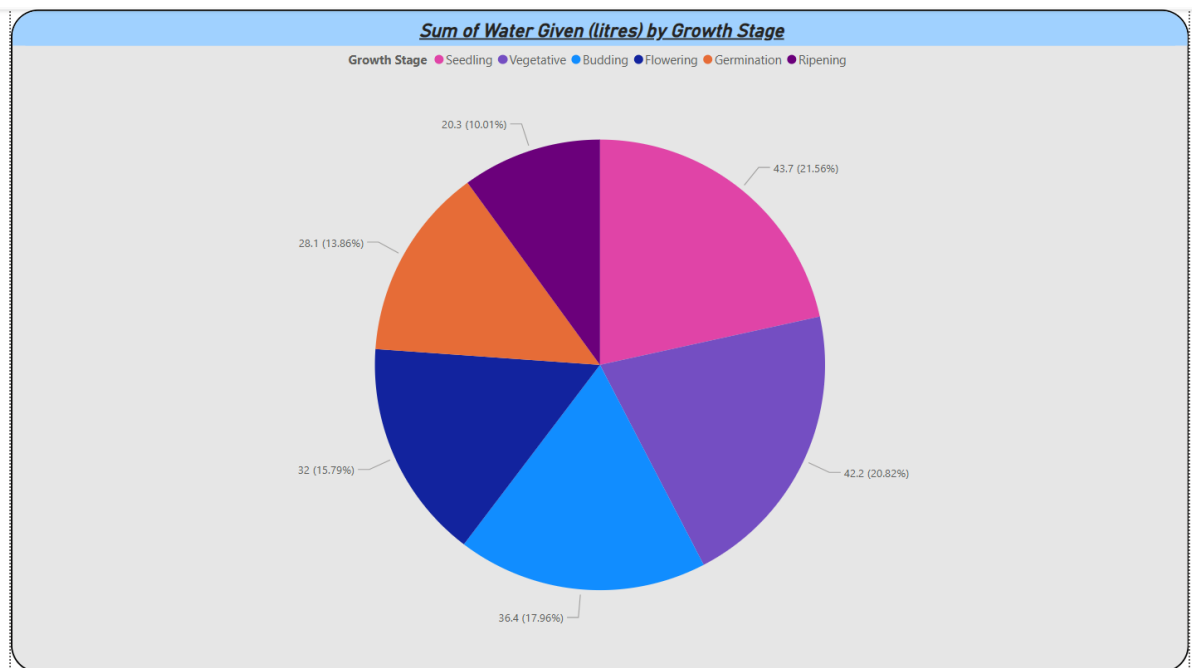
1. How does the sum of temperature vary across different growth stages?

- **Visualization:** Line chart showing temperature by growth stage.
- **Screenshot of visualisation**



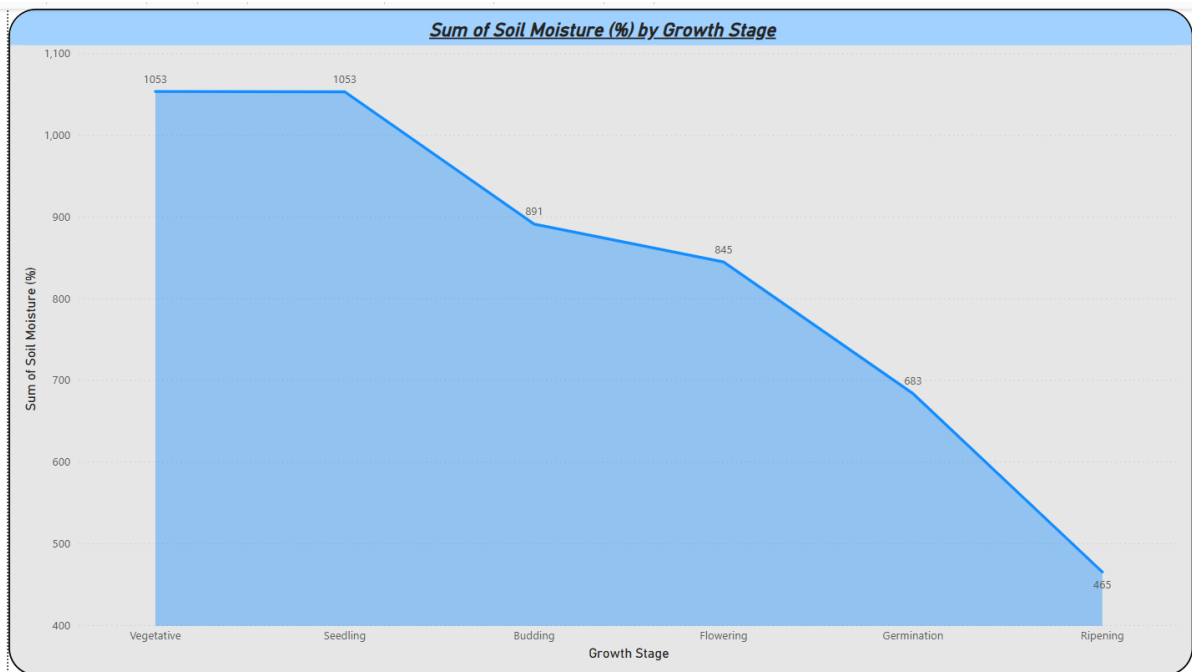
2. What is sum of water given to plants by their growth stages?

- **Visualization:** Pie Chart showing water given in litres to plants in various growth stages
- **Screenshot of visualisation**



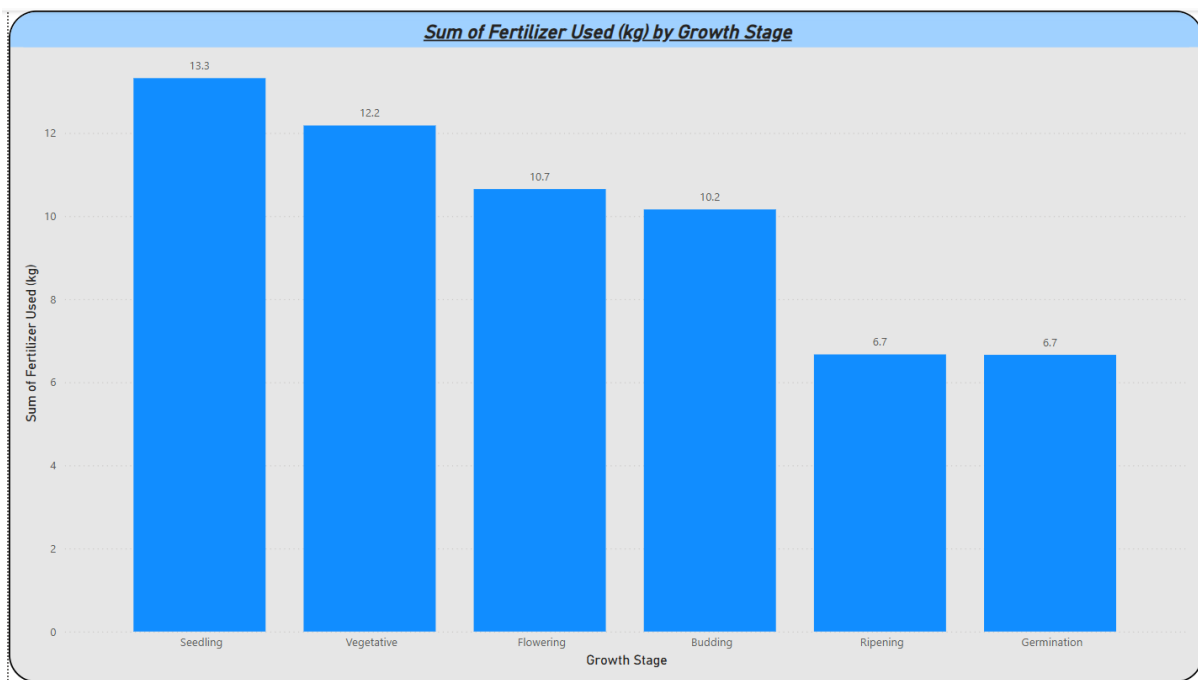
3. What is the correlation between soil moisture and growth stages?

- **Visualization:** Area Chart showing sum of soil moisture by their growth stage
- **Screenshot of visualisation**



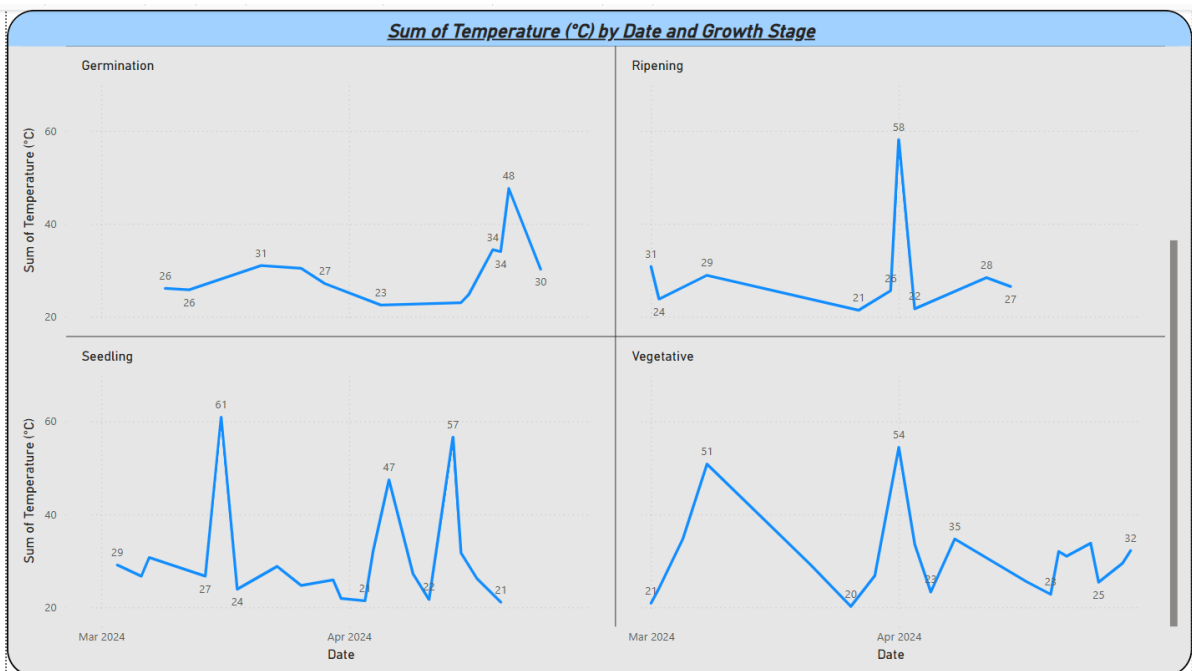
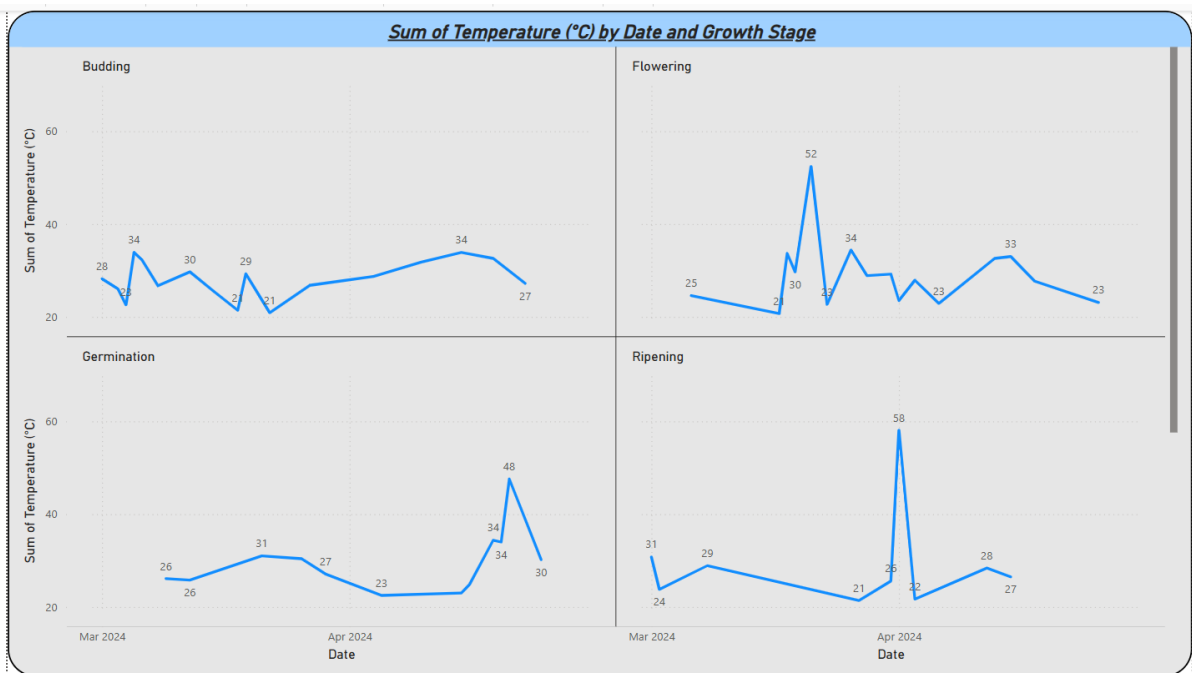
4. Which growth stage receives the highest (Sum) amount of fertiliser?

- **Visualization:** Stacked Column Chart showing fertiliser used by their growth stage
- **Screenshot of visualisation:**



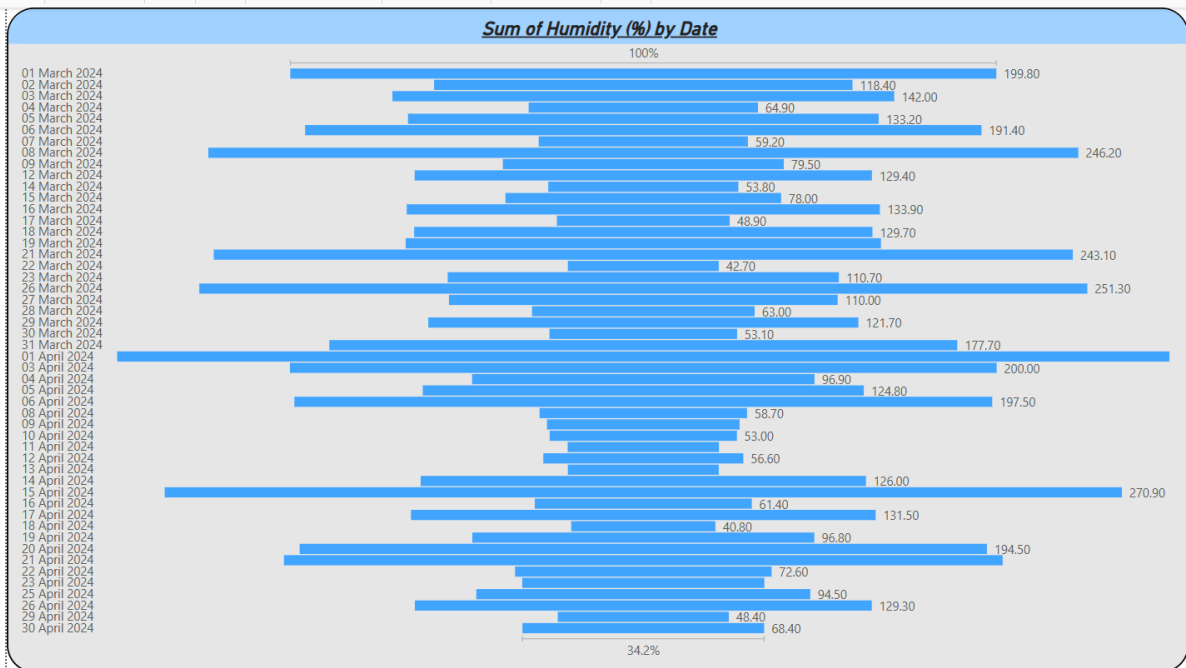
5. What is correlation between temperature and growth stage of plant?

- **Visualization:** Below are 2 charts showing sum of temperature by date and growth stage
- **Screenshot of visualisation:**



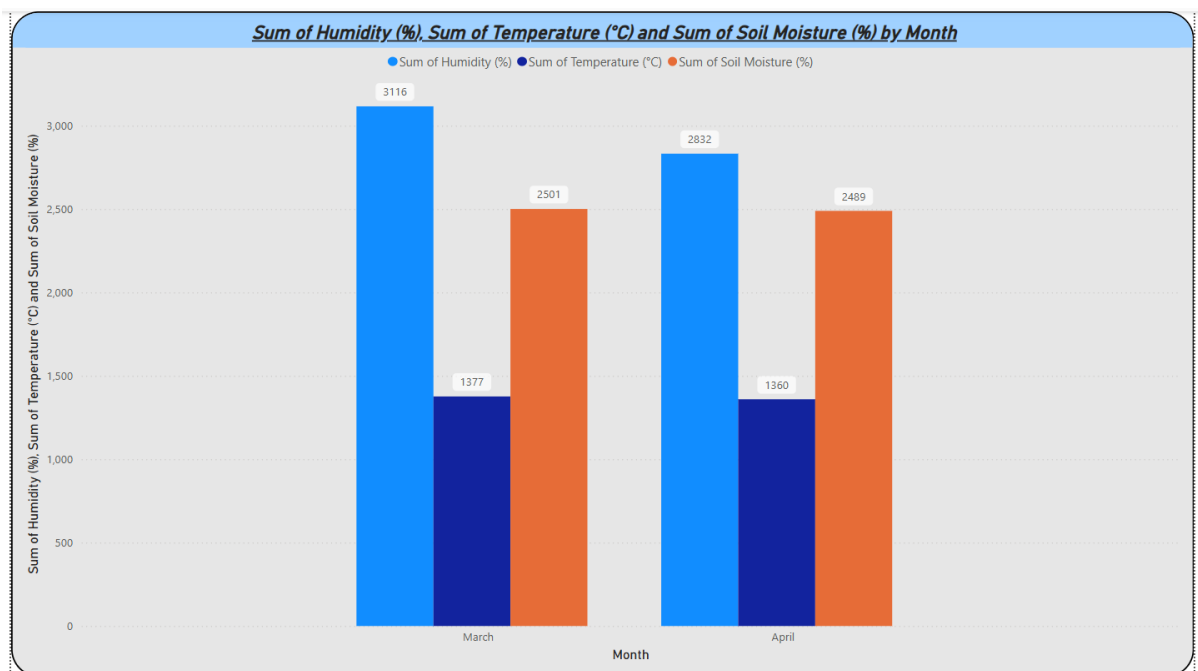
6. What is sum of humidity(%) on various different date?

- **Visualization:** This chart shows sum of humidity on various different date
- **Screenshot of visualisation:**



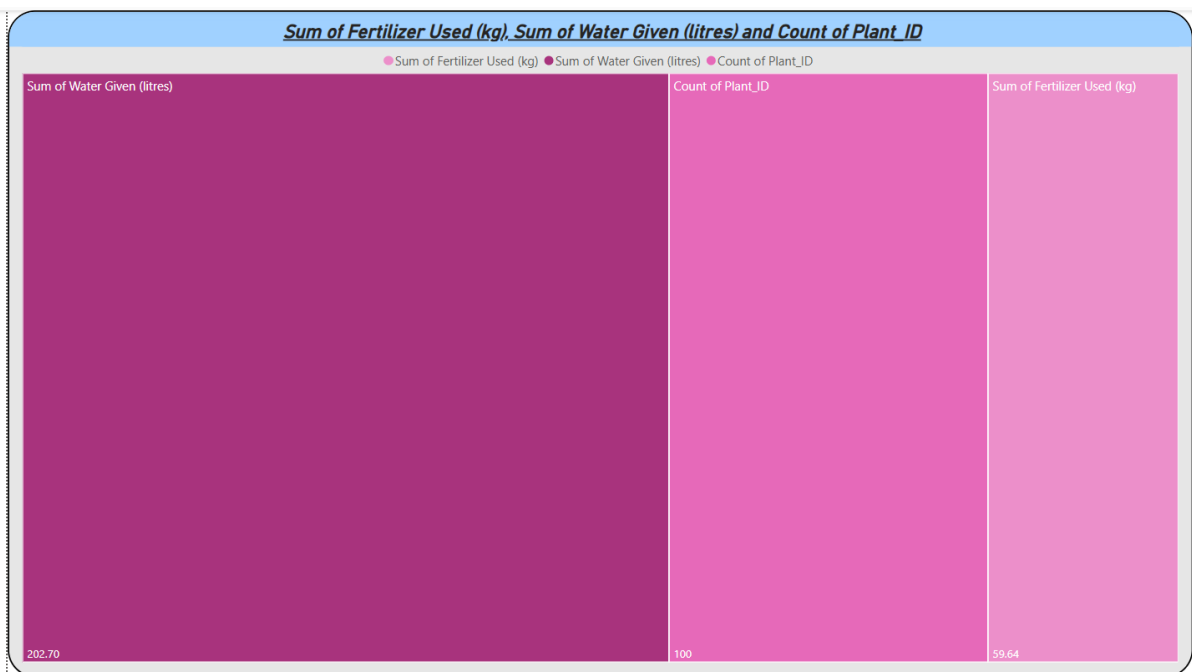
7. What is sum of humidity (%), temperature, and moisture (%) in various months?

- **Visualization:** This Clustered Column Chart shows sum of humidity (%), temperature, and moisture (%) in various months
- **Screenshot of visualisation:**



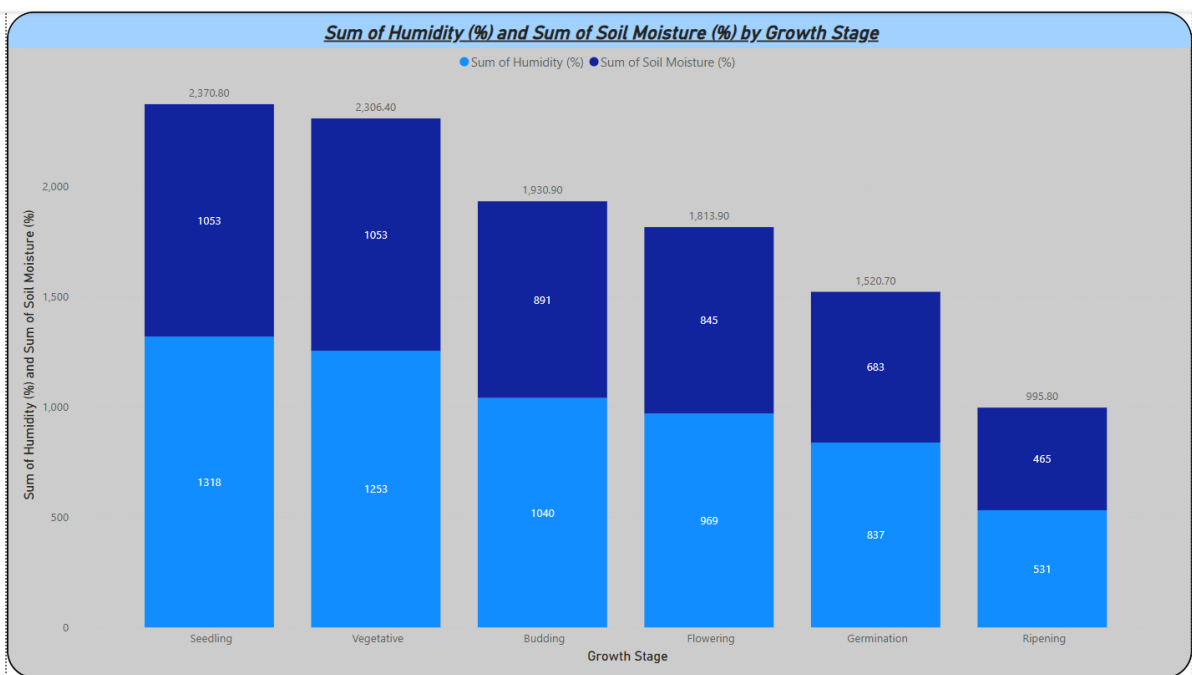
8. What is sum of fertiliser and water used as per their plant count ID?

- **Visualization:** This Treemap shows sum of fertiliser and water used as per their plant count ID
- **Screenshot of visualisation:**



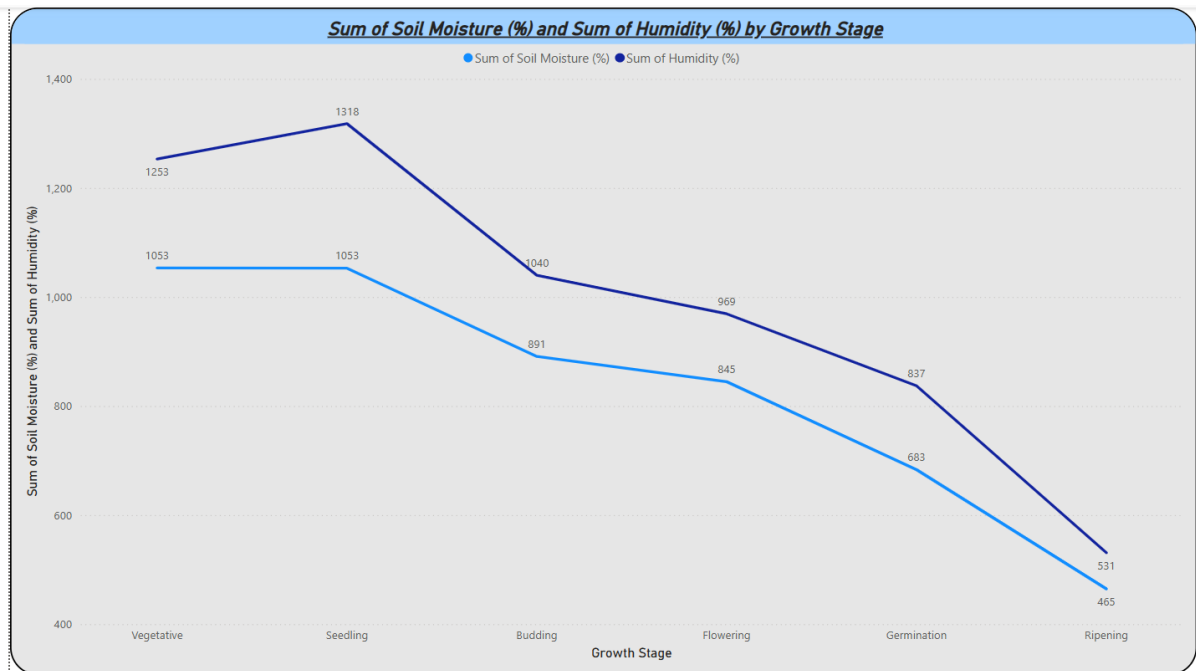
9. What is correlation between humidity and soil moisture as per their growth stage?

- **Visualization:** This Stacked Column Chart shows correlation between humidity and soil moisture as per their growth stage
- **Screenshot of visualisation:**



10. What is sum of soil moisture and sum of humidity is as per their different stages?

- **Visualization:** This Line Chart shows soil moisture and sum of humidity is as per their different stages.
- **Screenshot of visualisation**



Dashboard

Activity :

Interactive and visually appealing dashboards

To design a dashboard that is engaging and insightful, the following best practices were implemented:

- **Clear and Intuitive Layout:** Visuals are arranged logically with proper grouping (environmental insights on one side, growth stages on the other).
- **Use of Appropriate Visualizations:** Line charts, bar charts, pie charts, scatter plots, and combo charts were selected based on the type of data being analyzed.
- **Colour and Theming:** A consistent theme with high-medium-low color gradients (e.g., blue for high humidity, orange for low) enhances clarity.
- **Interactive Filters and Slicers:** Slicers were added for filtering data by growth stage and date for targeted analysis.
- **Drill-Down Capabilities:** Enabled on visuals like humidity trend and temperature analysis to explore detailed daily trends.

- **Responsive Design:** Visuals auto-adjust based on screen resolution and layout structure.
- **Custom Visuals and Icons:** Infographics and icons were used to visually represent growth stages.
- **Use of Infographics:** Symbols for fertilizer, water, and plant stages were used to make the dashboard visually engaging.



➤ Highlighted Dashboard Outcomes

- **Growth Stage Distribution:** A pie chart shows that "Vegetative" and "Ripening" stages have the highest record counts, indicating key development periods.
- **Average Temperature by Growth Stage:** A line chart reveals that temperature increases consistently until the flowering stage, then drops slightly in ripening.
- **Water and Fertilizer Usage Analysis:** Bar charts help identify that the "Seedling" and "Vegetative" stages require the most water and fertilizer combined.
- **Humidity Trends Over Time:** A date-wise line graph shows fluctuations, with peaks aligning to growth-sensitive stages.
- **Environmental Factor Comparison:** A combo chart comparing temperature, humidity, and soil moisture shows clear correlations with different growth stages.
- **Fertilizer vs Water Correlation:** A scatter plot with a trendline indicates a moderate positive relationship between water and fertilizer usage.

- **Stage-Based Recommendations:** Based on the interactive dashboard, actionable insights are derived for each growth stage for optimal input planning.
- **Dashboard Filtering:** Filters allow analysis of data by specific date ranges and growth stages, enabling targeted planning.

Report

A report is a comprehensive document that provides a detailed and structured account of data analysis, findings, and insights. It is typically used for in-depth analysis, documentation, and communication of results. Reports are suitable for a diverse audience, including decision-makers, analysts, and stakeholders who need a comprehensive understanding of the data.



Key Observations from the Power BI Report

1. Growth Stage Trends:

- The majority of the recorded plants were in the "Vegetative" and "Ripening" stages, showing they are dominant in the growth lifecycle.
- Average temperature peaked at the "Flowering" stage, while soil moisture was highest during the "Germination" stage.

2. Environmental Influence:

- A clear correlation was found between high humidity and early growth stages.
- Temperature showed a positive effect on stage progression until the ripening phase, beyond which it became less relevant.

3. Resource Usage Analysis:

- Fertilizer and water usage were highest during the "Vegetative" stage, indicating it requires the most input.
- The scatter plot between fertilizer and water usage showed a moderate positive correlation.

4. Time-Based Patterns:

- Humidity and temperature trends fluctuated over time but aligned closely with certain stages like "Budding" and "Flowering".
- Peak environmental conditions matched the timeline of active growth phases.

5. Performance Insights:

- Custom visuals allowed easy filtering of stages and helped in identifying best environmental conditions for each.
 - Region-wise or plant ID-specific filtering allowed granular insight into specific plant behaviors.
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Recommendation

- The Power BI report effectively links plant development to surrounding environmental and operational factors.
- Farmers and agricultural decision-makers can use this report to optimize fertilizer and water schedules.
- Future work can include real-time integration of IoT sensors to make the dashboard dynamic and predictive.

Performance Testing

7.1 Utilization of Data Filters

- Multiple slicers were implemented to filter data based on:
 - Growth Stage
 - Date
 - Environmental parameters (Temperature, Humidity, Soil Moisture)
- These filters enhance interactivity and allow users to view stage-specific trends, improving usability and performance clarity.

7.2 Number of Calculation Fields

- A total of 6 calculated fields (measures and columns) were created using DAX and Power Query:
 - Average Temperature
 - Average Humidity
 - Average Soil Moisture
 - Total Water Usage
 - Total Fertilizer Used
 - Management Effort Score (combined water and fertilizer)

7.3 Number of Visualizations

- The report includes 8 interactive visualizations:
 1. Average Temperature by Growth Stage (Line Chart)

2. Plant Distribution across Growth Stages (Pie Chart)
3. Soil Moisture vs Growth Stage (Scatter Plot)
4. Average Fertilizer Used (Bar Chart)
5. Water Usage per Stage (Column Chart)
6. Humidity Trend Over Time (Line Chart)
7. Environmental Factors Comparison (Combo Chart)
8. Fertilizer vs Water Correlation (Scatter Plot with Trendline)

Conclusion / Observation

The Power BI-based analysis of plant growth stages using environmental and management data provided valuable insights into how temperature, humidity, soil moisture, water input, and fertilizer usage influence plant development.

The visualizations clearly showed that different stages of plant growth respond uniquely to environmental factors. For example, high humidity benefits early growth stages like germination, while stages like vegetative and flowering require more water and fertilizer. These patterns were effectively captured through interactive visuals and filters.

The report confirms that data-driven farming decisions can enhance productivity, optimize input usage, and help forecast crop development stages. The dashboard's user-friendly design and analytical depth support precision agriculture and demonstrate how data science can empower sustainable farming practices.

This project successfully met its objective of linking environmental and operational data to plant development, creating a practical and scalable decision-support tool.

Future Scope

The current project demonstrates how environmental and management data can be used to predict plant growth stages effectively. However, there are several promising directions for future enhancement:

Data-Related Improvements:

- **Real-Time Sensor Integration:** Future datasets can be enriched using IoT sensors (e.g., soil probes, weather stations) for live temperature, humidity, and moisture updates.
- **Multiyear & Multilocation Data:** Expanding the dataset across multiple seasons and regions will improve model robustness and generalization.
- **Disease & Pest Data:** Adding data related to pest infestations or disease occurrence can enable predictive plant health analysis.
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Power BI Utilization Expansion:

- **Advanced Predictive Visuals:** Integrate machine learning models outside Power BI and import predictions via DAX or APIs for real-time forecasts.
- **Geo-Mapping Capabilities:** Use map visuals in Power BI to show crop conditions by region or farm.
- **Alert-Based Dashboards:** Add thresholds and KPIs to trigger visual alerts when water/fertilizer levels cross safe limits.

- **Mobile Dashboard Access:** Optimize layout for field-access via mobile Power BI apps, enabling real-time use by farmers and agronomists.

10. Appendix

10.1 Source

The following data sources were used for analysis, modeling, and visualization in this project:

Source Name	Description	URL / Location	Access Type
Dataset 1: Plant Growth Data Classification	Contains plant growth stages with environmental factors like temperature, humidity, and soil moisture.	https://www.kaggle.com/datasets/gorororororo23/plant-growth-data-classification	Public
Dataset 2: Greenhouse Plant Growth Metrics	Includes growth metrics, biomass data, and chlorophyll levels across 30,000+ plant samples.	https://www.kaggle.com/datasets/adilshamim8/greenhouse-plant-growth-metrics	Public
Dataset 3: Custom Synthetic Dataset	Designed by the project team to simulate realistic environmental and management data.	Locally Generated (Excel format)	Private

